

HOMEWORK 6

INTRODUCTION TO COMPLEX ANALYSIS

(MATH 4230-40), SPRING 2022

SECTION 2.3 - A CLOSER LOOK AT CAUCHY'S THEOREM

1. Prove that $\mathbb{C} \setminus \{0\}$ is not simply connected.

Recall that a connected set G is said to be *simply connected* if every closed curve γ in G is homotopic (as a closed curve) to a point in G . So, we must show that there exists a closed curve in $\mathbb{C} \setminus \{0\}$ which is not homotopic as a closed curve to any point in $\mathbb{C} \setminus \{0\}$. To do this, recall that the homotopy form of Cauchy's theorem (i.e. Theorem 2.3.14) tells us that if γ is a curve in $\mathbb{C} \setminus \{0\}$ which is homotopic to a point in $\mathbb{C} \setminus \{0\}$, then $\int_{\gamma} f = 0$ for any function f which is analytic on $\mathbb{C} \setminus \{0\}$. Rephrasing this in the contrapositive, this says that if f is analytic on $\mathbb{C} \setminus \{0\}$ and γ is a curve in $\mathbb{C} \setminus \{0\}$, then if $\int_{\gamma} f \neq 0$, γ is not homotopic to a point in $\mathbb{C} \setminus \{0\}$.

Take $f(z) = \frac{1}{z}$, which is analytic on $\mathbb{C} \setminus \{0\}$, and let $\gamma(t) = e^{2\pi it}$ for $t \in [0, 1]$. This gives

$$\begin{aligned}\int_{\gamma} f &= \int_0^1 \frac{1}{\gamma(t)} \gamma'(t) dt \\ &= \int_0^1 \frac{2\pi i e^{2\pi it}}{e^{2\pi it}} dt \\ &= 2\pi i.\end{aligned}$$

Therefore, $\int_{\gamma} f \neq 0$, and since f is analytic on $\mathbb{C} \setminus \{0\}$ and γ is a curve in $\mathbb{C} \setminus \{0\}$, it then follows from the contrapositive of the homotopy form of Cauchy's theorem that γ is not homotopic to a point in $\mathbb{C} \setminus \{0\}$. Accordingly, $\mathbb{C} \setminus \{0\}$ is not simply connected.

2. Show that every disk is convex.

Let $D(z_0, R)$ denote the disk centered at $z_0 \in \mathbb{C}$ of radius R . Note that $z \in D(z_0, R)$ may be written as $z = z_0 + re^{i\theta}$ for some $\theta \in [0, 2\pi)$ and $r \in [0, R]$. Let $z_1 = z_0 + r_1 e^{i\theta_1}$ and $z_2 = z_0 + r_2 e^{i\theta_2}$ be two points in $D(z_0, R)$. To show that $D(z_0, R)$ is convex, we must show that $tz_1 + (1-t)z_2 \in D(z_0, R)$ for all $t \in [0, 1]$. Since

$$\begin{aligned} (tz_1 + (1-t)z_2) &= tz_0 + tr_1 e^{i\theta_1} + (1-t)z_0 + (1-t)r_2 e^{i\theta_2} \\ &= z_0 + (tr_1 e^{i\theta_1} + (1-t)r_2 e^{i\theta_2}), \end{aligned}$$

this is equivalent to showing that $(tr_1 e^{i\theta_1} + (1-t)r_2 e^{i\theta_2}) \in D(0, R)$. So, notice that

$$\begin{aligned} |tr_1 e^{i\theta_1} + (1-t)r_2 e^{i\theta_2}| &\leq |tr_1 e^{i\theta_1}| + |(1-t)r_2 e^{i\theta_2}| \quad (\text{By the triangle inequality, } |a+b| \leq |a| + |b|) \\ &= t|r_1 e^{i\theta_1}| + (1-t)|r_2 e^{i\theta_2}| \quad (\text{Since } |cz| = c|z| \text{ for any } 0 \leq c \in \mathbb{R}) \\ &= t|r_1| |e^{i\theta_1}| + (1-t)|r_2| |e^{i\theta_2}| \quad (\text{Since } |zw| = |z| |w|) \\ &= t|r_1| + (1-t)|r_2| \quad (\text{As } |e^{i\theta}| = 1 \text{ for all values } \theta \in \mathbb{R}) \\ &\leq tR + (1-t)R \quad (\text{Since } r_1, r_2 \leq R) \\ &= R. \end{aligned}$$

Hence, $(tr_1 e^{i\theta_1} + (1-t)r_2 e^{i\theta_2}) \in D(0, R)$, and it then follows that

$$(tr_1 e^{i\theta_1} + (1-t)r_2 e^{i\theta_2}) + z_0 = (tz_1 + (1-t)z_2) \in D(z_0, R).$$

Therefore, $D(z_0, R)$ is convex.

3. Evaluate the following integrals without performing explicit computation:

(a) $\int_{\gamma} \frac{dz}{z}$, where $\gamma(t) = \cos t + 2i \sin t$, $0 \leq t \leq 2\pi$.

Note that γ is a parameterization of an ellipse, which is homotopic as a closed curve to the unit circle. Accordingly, $\int_{\gamma} \frac{dz}{z} = 2\pi i$.

(b) $\int_{\gamma} \frac{dz}{z^2}$, where $\gamma(t) = \cos t + 2i \sin t$, $0 \leq t \leq 2\pi$.

Since $\frac{1}{z^2}$ has an antiderivative on $\mathbb{C} \setminus \{0\}$ and γ is a closed curve lying in this region, $\int_{\gamma} \frac{dz}{z^2} = 0$.

(c) $\int_{\gamma} \frac{e^z dz}{z}$, where $\gamma(t) = 2 + e^{it}$, $0 \leq t \leq 2\pi$.

Note that γ is a parameterization of a circle of radius 1 centered at 2. Since $\frac{e^z}{z}$ is analytic on $\mathbb{C} \setminus \{0\}$ and γ is a closed curve in this region, $\int_{\gamma} \frac{e^z dz}{z} = 0$.